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CASE RECORDS of the MASSACHUSETTS GENERAL HOSPITAL

Case 31-2022: A 72-Year-Old Man with Heartburn, Nausea, and Inability to Eat

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PRESENTATION OF CASE

Dr. Yousef R. Badran: A 72-year-old man was transferred to this hospital because of heartburn, nausea, and the inability to eat solid and liquid food.

The patient had a history of gastroesophageal reflux disease. The symptoms had been well controlled with the use of daily omeprazole therapy until 11 months before this admission, when heartburn recurred despite treatment. At that time, the patient also noticed nausea after eating and early satiation. During the next 9 months, the heartburn and nausea slowly increased in severity. To help manage his symptoms, the patient adjusted his diet from solid food to soft solid food.

Two months before this admission, the patient sought evaluation at the gastroenterology clinic of another hospital. The glycated hemoglobin level was 6.1% (reference range, 4.3 to 5.6). Esophagogastroduodenoscopy (EGD) was performed. Although the patient had not eaten food for 24 hours before the procedure, the stomach could not be adequately visualized because there was a large amount of residual food; no intraluminal masses were seen.

A scintigraphy study of gastric emptying was performed. During the study, 88% of the gastric contents were retained at 1 hour, 88% at 2 hours, 81% at 3 hours, and 80% at 4 hours. In a patient with normal gastric emptying, less than 60% of the gastric contents would be retained at 2 hours and less than 10% at 4 hours. Thus, the results were consistent with severely delayed gastric emptying. The patient was told that he had idiopathic gastroparesis and underwent trials of several medications. Ondansetron caused constipation, erythromycin caused burning while swallowing and abdominal cramping, and metoclopramide and scopolamine did not lead to a decrease in the patient's symptoms.

During the 2 months after the evaluation in the gastroenterology clinic, the patient adjusted his diet from soft solid food to blended solid food. Two weeks before this admission, he adjusted his diet to primarily liquid food, such as nutritional supplement drinks, protein shakes, and ice cream. When the patient was no longer able to drink liquids, he sought evaluation at the emergency department of the other hospital.

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On evaluation, the patient described epigastric burning and nausea after eating. He had lost 13.6 kg of weight in the preceding year, including 5.4 kg in the past 2 weeks. He had no recent history of illness, night sweats, vomiting, dysphagia, odynophagia, abdominal pain, or diarrhea.

Dr. David A. Rosman: Computed tomography (CT) of the chest, abdomen, and pelvis, performed after the administration of intravenous contrast material, revealed a patulous esophagus, along with wall thickening in scattered areas of the upper and lower segments of the esophagus. There was abnormal circumferential wall thickening (≤13 mm) in the stomach, particularly at the antrum, incisura, and pyloric canal (Fig. 1). The antrum was decompressed or nonexpansile, and the rest of the stomach was distended. There were multiple bilateral pulmonary nodules (≤5 mm in diameter) and multiple hepatic hypodensities (<10 mm in diameter), as well as a pancreatic cyst (5 mm in diameter) and an exophytic renal lesion (8 mm in diameter).

Dr. Badran: A nasogastric tube was inserted. Treatment with intravenous pantoprazole was started, and intravenous lorazepam was administered for nausea. The patient was admitted to the other hospital.

During the subsequent week, the patient received vitamin and mineral supplements for severe malnutrition. He remained unable to drink liquids. On the 7th hospital day, a peripherally inserted central catheter was placed, and total parenteral nutrition was administered. The patient was monitored for refeeding syndrome. On the 15th hospital day, he was transferred to this hospital for further treatment.

On transfer to this hospital, additional history was obtained. The patient had a history of prediabetes, hypertension, dyslipidemia, diverticulosis, glaucoma, and basal cell carcinoma of the ear that had been resected. Five years before this admission, EGD had revealed a widely patent Schatzki's ring, an irregular-appearing Z line, and a few gastric polyps; examination of an esophageal biopsy specimen had revealed no abnormalities. Colonoscopy had shown two tubular adenomas. The patient had undergone orthognathic surgery, but there was no history of abdominal surgery. Medications included omeprazole, lisinopril, atorvastatin, vitamin D supplements, and bimatoprost and bromonidine eye drops. Sulfa drugs had caused a rash.

The patient lived with his wife in a rural area of New England. He had previously served in the military and currently worked as a biotechnologist. He had never smoked tobacco, and he drank alcohol rarely. His mother and paternal aunt had died of colon cancer, his father of lung cancer, his maternal grandmother of breast cancer, and his maternal grandfather of prostate cancer. His two children and two grandchildren were healthy.

On examination, the temperature was 37.7°C, the blood pressure 128/76 mm Hg, the pulse 105 beats per minute, the respiratory rate 18 breaths per minute, and the oxygen saturation 97% while the patient was breathing ambient air. He was thin and had a nasogastric tube in place. He had normal mood, affect, and insight. The abdomen was flat, soft, and nondistended, with mild tenderness in the epigastrium on palpation. There was no palpable lymphadenopathy. Distal sensory and motor function were normal. Laboratory test results are shown in Table 1.

A diagnostic test was performed.

DIFFERENTIAL DIAGNOSIS

Dr. Kyle D. Staller: This 72-year-old man presented with heartburn and nausea with the progressive inability to tolerate solids and liquids. How can the patient's symptoms help us understand the underlying process?

HEARTBURN

The predominant mechanism of gastroesophageal reflux is thought to be transient relaxation of the lower esophageal sphincter, but there are other mechanisms, including impaired esophageal clearance, anatomical disruption of the lower esophageal sphincter complex, and delayed gastric emptying. This patient was known to have retention of gastric contents, and such retention can increase the likelihood of reflux and associated heartburn. In addition, he had gastric-wall thickening, which could suggest disruption of the intricate network of clasplike sling fibers that originate from the upper stomach and form part of the normal barrier to reflux.

EARLY SATIATION

Satiation is a feeling of fullness that occurs during a meal and controls the size of the meal. Satiation is not to be confused with satiety,

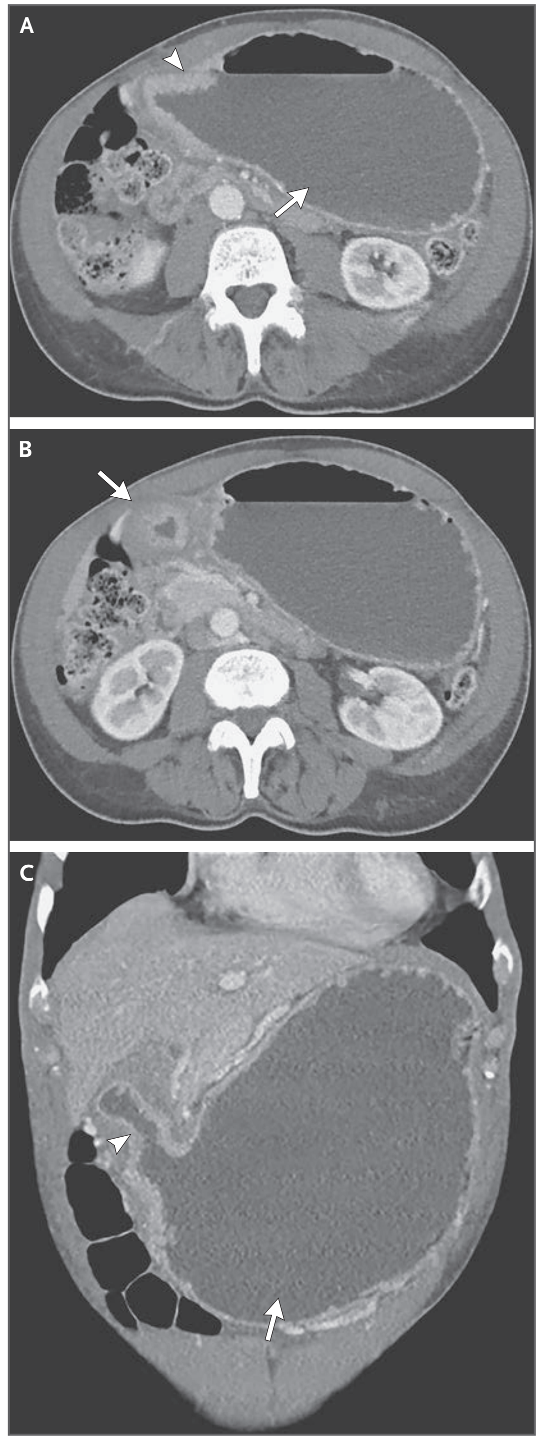

Figure 1. CT Scan of the Abdomen.
CT was performed after the administration of intravenous contrast material. An axial image (Panel A) shows wall thickening at the antrum (arrowhead) and secondary distention of the stomach (arrow). An additional axial image (Panel B) shows circumferential wall thickening at the incisura and pyloric canal (arrowhead). A coronal image (Panel C) shows the antral thickening (arrowhead) and gastric distention (arrow).

which is a feeling of fullness that occurs after a meal and controls postmeal hunger. This patient had noticed early satiation, and he was found to have delayed gastric emptying. However, gastric emptying is just one aspect of the complex physiology that governs food intake. Another aspect — and the one that is most relevant to early satiation — is gastric accommodation. Gastric accommodation is a vagally mediated, adaptive process that consists of receptive relaxation of the proximal stomach (fundus) in response to food intake, which provides a reservoir for incoming food without increasing intragastric pressure (Fig. 2A). When gastric accommodation is impaired, ingested food redistributes to other parts of the stomach, which increases intragastric pressure and triggers early satiation (Fig. 2B).

FOOD INTOLERANCE

This patient had progressive food intolerance, in which he adjusted his diet from solids to soft solids and finally to liquids before becoming unable to eat altogether. The finding of delayed gastric emptying is relevant to progressive food intolerance. Gastric emptying is a neurohormonally controlled process that occurs after gastric accommodation, in which digestible solids are pulverized in the stomach into small particles and then emptied into the duodenum through coordinated antral contractions (Fig. 2C). Gastric emptying is highly dependent on food characteristics, including volume, caloric density, and chemical composition.[1] Low-calorie liquids are emptied from the stomach passively, whereas high-calorie liquids, digestible solids, and indigestible solids require progressively more gastric effort. When gastric emptying is delayed, the emptying of high-fat foods and indigestible solids is affected first; retention of gastric contents correlates with symptoms such as nausea, vomiting, and abdominal pain (Fig. 2D). Gastroparesis is diagnosed when these symptoms correlate with evidence of delayed gastric emptying on scintigraphy.

GASTROPARESIS

The presence or degree of delayed gastric emptying has not been shown to correlate with symp-

Table 1. Laboratory Data.*

Variable	Reference Range, Adults, This Hospital†	On Transfer, This Hospital
Sodium (mmol/liter)	135–136	138
Potassium (mmol/liter)	3.4–5.0	3.4
Chloride (mmol/liter)	98–108	93
Carbon dioxide (mmol/liter)	23–32	33
Urea nitrogen (mg/dl)	8–25	27
Creatinine (mg/dl)	0.60–1.50	0.93
Glucose (mg/dl)	70–110	110
Calcium (mg/dl)	8.5–10.5	9.6
Magnesium (mg/dl)	1.7–2.4	2.1
Phosphorus (mg/dl)	2.6–4.5	3.3
Prealbumin (mg/dl)	20–40	22
Albumin (g/dl)	3.3–5.0	3.8
Globulin (g/dl)	1.9–4.1	3.6
Total protein (g/dl)	6.0–8.3	7.4
Direct bilirubin (mg/dl)	0.0–0.4	0.2
Total bilirubin (mg/dl)	0.0–1.0	0.5
Alanine aminotransferase (U/liter)	10–55	25
Aspartate aminotransferase (U/liter)	10–40	19
Alkaline phosphatase (U/liter)	45–115	86
Glycated hemoglobin (%)	4.3–5.6	6.7
Hemoglobin (g/dl)	13.5–17.4	14.4
Hematocrit (%)	41.0–53.0	42.8
Platelet count (per μl)	150,000–400,000	197,000
White-cell count (per μl)	4500–11,000	7890

* To convert the values for urea nitrogen to millimoles per liter, multiply by 0.357. To convert the values for creatinine to micromoles per liter, multiply by 88.4. To convert the values for glucose to millimoles per liter, multiply by 0.05551. To convert the values for calcium to millimoles per liter, multiply by 0.250. To convert the values for magnesium to millimoles per liter, multiply by 0.4114. To convert the values for phosphorus to millimoles per liter, multiply by 0.3229. To convert the values for bilirubin to micromoles per liter, multiply by 17.1.

† Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

tom severity.[2] Nevertheless, retention of 80% of gastric contents at 4 hours during a scintigraphy study, a finding that was observed in this patient, is indicative of severely delayed gastric emptying. Acute hyperglycemia or poor long-term glycemic control can place patients at risk for gastroparesis, but this patient's glycated hemoglobin level was only mildly elevated, and his symptom progression was more rapid than would be expected with a typical case of diabetic gastroparesis. Such rapid progression may be seen with paraneoplastic gastrointestinal dysmotility, a condition that is often associated with the presence of antineuronal nuclear antibodies. These antibodies have been associated with small-cell lung cancer and thymoma, and the pulmonary lesions observed in this patient could suggest these diagnoses. However, because low-calorie liquids are emptied from the stomach passively, the patient's subsequent inability to tolerate fluids suggests some degree of gastric-outlet obstruction, and gastroparesis is unlikely to explain his presentation.

INFILTRATIVE PROCESSES

Which disease processes can cause gastric-wall thickening, impaired gastric accommodation, and gastric-outlet obstruction? The stomach lining consists of the following layers: mucosa (the inner layer), submucosa, muscularis propria, and serosa. The muscularis propria layer contains oblique, circular, and longitudinal muscle. Embedded within the muscularis propria layer is the myenteric plexus, the primary neurologic control system for gastric emptying. This patient's progressive symptoms, gastric-wall thickening, impaired gastric accommodation, and gastric-outlet obstruction suggest that an infiltrative process affecting the muscularis propria layer is the most likely cause of his symptoms.

Amyloidosis

Several infiltrative disorders can affect the stomach. Gastric amyloidosis results in the deposition of insoluble fibers in the stomach. Most cases of gastric amyloidosis are part of a systemic disease, and involvement of the gastric body is more common than involvement of the antrum. This patient had wall thickening predominantly in the antrum, and he did not have signs of systemic amyloidosis. In addition, many patients with gastric amyloidosis are asymptomatic, and the disease is more likely to involve the mucosal and submucosal layers than the muscularis propria layer.[3]

Lymphoma

Gastric lymphoma is another important consideration in this patient. The stomach is the most

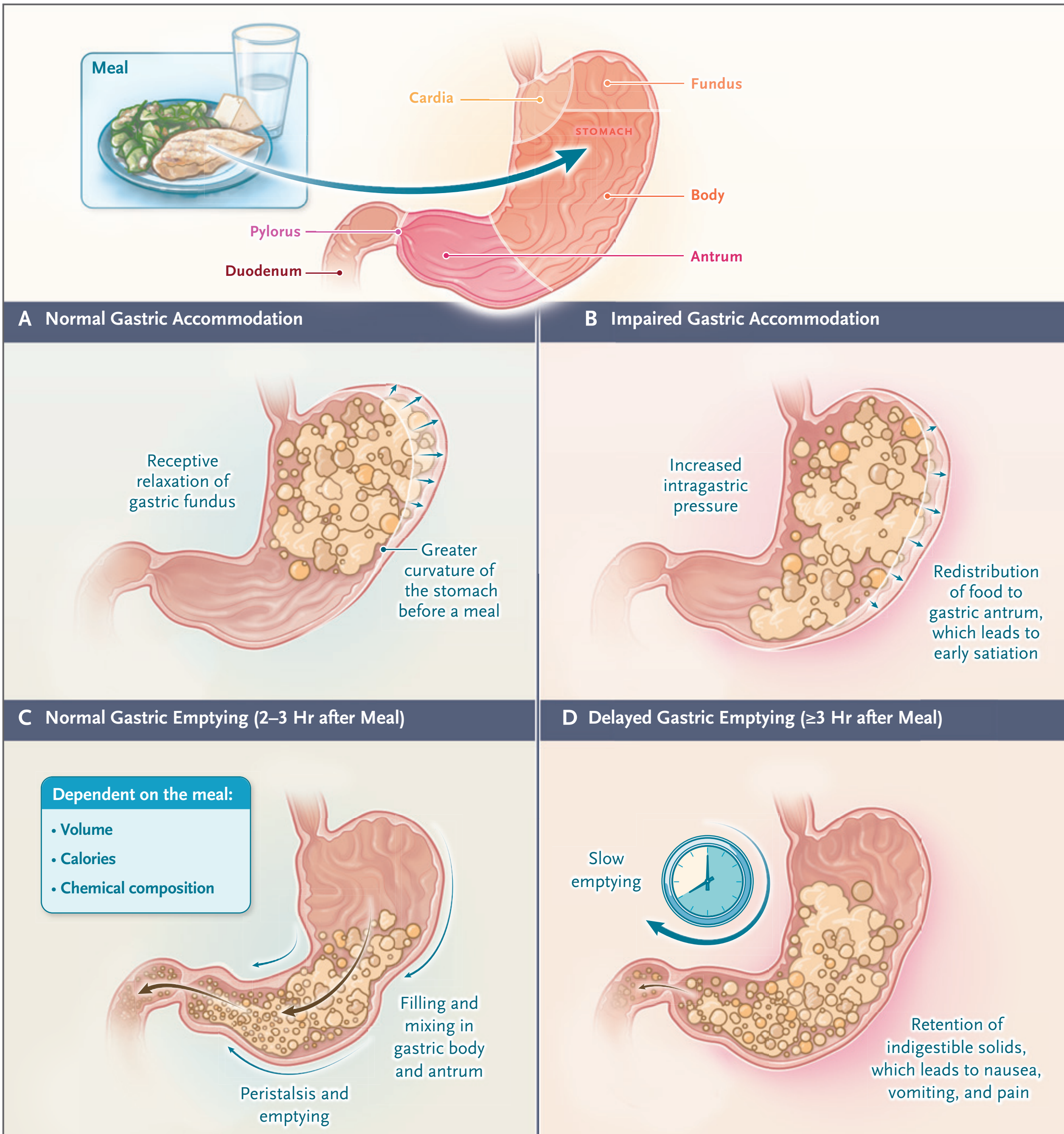

Figure 2. Gastric Response to a Meal.

Gastric accommodation is a process that consists of receptive relaxation of the proximal stomach (fundus) in response to food intake, which allows for increased volume intake without increased intragastric pressure (Panel A). When gastric accommodation is impaired, ingested food redistributes to other parts of the stomach, which increases intragastric pressure and triggers early satiation (Panel B). Gastric emptying is a process that occurs after gastric accommodation, in which digestible solids are pulverized in the stomach into small particles and then emptied into the duodenum through coordinated antral contractions (Panel C). When gastric emptying is delayed, retention of gastric contents (especially high-fat foods and indigestible solids) correlates with symptoms such as nausea, vomiting, and abdominal pain (Panel D).

common site of extranodal lymphoma. In patients with gastric lymphoma, imaging frequently shows diffuse gastric-wall thickening. However, luminal pliability is typically preserved, lymphadenopathy is common, and gastric-outlet obstruction is rare.[4]

Eosinophilic Gastroenteritis

Eosinophilic gastroenteritis is an increasingly recognized entity that is characterized by eosinophilic infiltration of the gastrointestinal tract. Eosinophilic gastroenteritis can affect any layer of the gastrointestinal lining, including the muscularis propria layer. Antral wall thickening can also occur and may result in gastric-outlet obstruction. However, eosinophilic gastroenteritis is most commonly manifested by mucosal disease, and most cases that involve the muscularis propria layer are associated with peripheral eosinophilia.[5]

Sarcoidosis

Sarcoidosis is a systemic disease that is identified by the presence of noncaseating granulomas in involved tissues. The stomach is the most frequently involved gastrointestinal luminal organ; when the liver is involved, multiple hypodense liver lesions are often detected on imaging.[6] It is important to note that 58% of patients with gastric sarcoidosis present with extensive, diffuse infiltration of the gastric wall.[7] However, gastric sarcoidosis typically develops in patients who are 20 to 50 years of age, so the age at onset would be atypical in this patient.

Gastric Adenocarcinoma

Gastric adenocarcinoma frequently takes the form of an ulcerated, discrete mass. However, a morphologic variant of diffuse gastric cancer — linitis plastica — can cause gastric-wall thickening. Linitis plastica is associated with scirrhous or fibrous stroma that results from abnormal growth of submucosal connective tissue. The abnormal stroma causes hypertrophy of the gastric muscularis propria and subserosal thickening.[8] These changes result in diffuse gastric-wall thickening that often starts at the antrum, but enlargement of the gastric folds can occur as well. These features are consistent with the imaging findings observed in this patient. Moreover, hypertrophy of the submucosal and muscularis propria layers of the gastric wall can impair both gastric accommodation and gastric emptying. Progressive disease can lead to gastric-outlet obstruction. Because the liver is the common venous drainage point for most of the gastrointestinal tract, it is a frequent site of hematogenous metastases, which appear as hypodense lesions on CT.

This patient initially had breakthrough heartburn that did not respond to previously effective acid-suppression therapy, which could represent malignant infiltration of the clasplike sling fibers that originate from the upper stomach and form part of the normal acid barrier. Malignant infiltration of the submucosal and muscularis propria layers of the fundus impairs normal gastric accommodation, which leads to early satiation. Malignant infiltration of the entire antral wall disrupts the neurocircuitry and muscular contractions that are required for gastric emptying. This process initially mimics gastroparesis, with retention of solid food and nausea, but prokinetic treatments are ineffective because damage to the muscularis propria prevents these agents from effectively triggering antral contractions. Liquids are initially tolerated because they are emptied passively. However, progressive antral disease marks a transition from gastroparesis to gastric-outlet obstruction, and liquids are no longer tolerated. Patients with linitis plastica frequently present with advanced disease, and this patient's hypodense liver lesions may suggest metastatic disease.

Given this patient's rapidly progressive symptoms that are suggestive of specific alterations in gastric physiology, in combination with his imaging and EGD findings, I favor a diagnosis of gastric cancer that is consistent with linitis plastica. The diagnostic test of choice would be endoscopic ultrasonography (EUS).

DR. KYLE D. STALLER'S DIAGNOSIS

Linitis plastica (invasive gastric adenocarcinoma).

DIAGNOSTIC TESTING

Dr. Badran: Upper-endoscopic evaluation revealed reflux esophagitis in the lower third of the esophagus. The severity was determined to be Los Angeles classification grade C; the grades range from A to D, with grade D indicating the most severe esophagitis.[9] Diffuse nodular mucosa

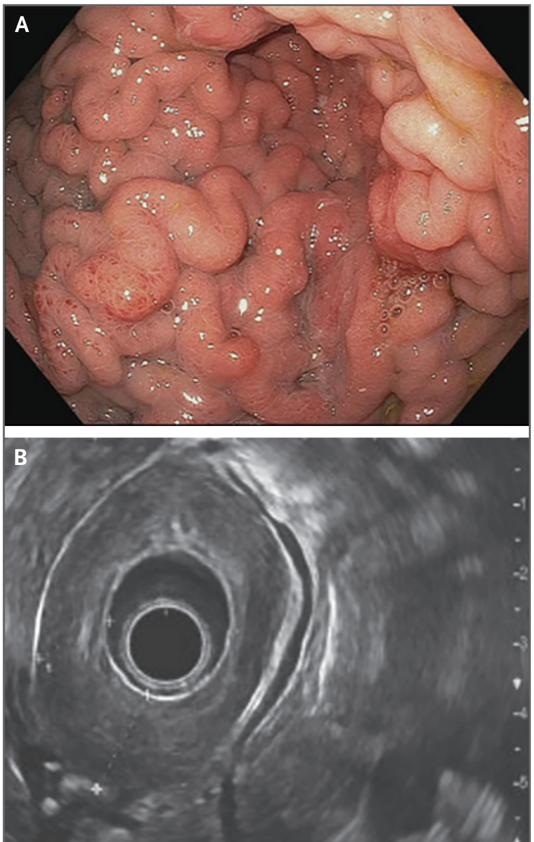

Figure 3. Upper-Endoscopic and Endoscopic Ultrasound Images.

An endoscopic image of the stomach (Panel A) shows diffuse nodular mucosa in the prepyloric region. An ultrasound image (Panel B) shows patchy wall thickening (≤15 mm) involving the deep mucosal, submucosal, and muscularis propria layers.

was found along the anterior wall of the gastric body and at the antrum (Fig. 3A). The examined duodenum was normal. EUS revealed patchy wall thickening (≤15 mm) in the lesser curvature, incisura, and antrum of the stomach. The thickening appeared to primarily involve the deep mucosal, submucosal, and muscularis propria layers (Fig. 3B). The appearance of the gastric wall was suggestive of an infiltrative neoplasm. Fine-needle biopsy was performed.

PATHOLOGICAL DISCUSSION

Dr. Richard Judelson: Histologic examination of the biopsy specimen from the stomach revealed fragments of muscularis propria with an associated infiltrative, poorly differentiated adenocarcinoma (Fig. 4A). The background foveolar epithelium was unremarkable. Ancillary studies showed preserved expression of mismatch-repair proteins, no expression of programmed death ligand 1 (PD-L1), and equivocal expression of human epidermal growth factor receptor 2 (HER2) on immunohistochemical staining, with no amplification seen on fluorescence in situ hybridization.

PATHOLOGICAL DIAGNOSIS

Invasive gastric adenocarcinoma.

DISCUSSION OF MANAGEMENT

Dr. Samuel J. Klempner: Although gastric cancer is uncommon in the United States, it is the fifth leading cause of cancer-specific death worldwide, with more than 768,000 deaths in 2020.[10] The main risk factors for gastric cancer are infection with the group 1 carcinogen *Helicobacter pylori* and a positive family history.[11] The diffuse gastric-cancer subtype, which includes linitis plastica, is often associated with truncating alterations in E-cadherin on chromosome 16 that lead to disrupted cell-to-cell adhesion; the diffuse subtype is less often associated with *H. pylori.* Unfortunately, most U.S. patients with gastric cancer present with locally advanced or metastatic disease. Unlike countries with a high prevalence of gastric cancer (e.g., South Korea and Japan), the United States does not offer routine gastric-cancer screening. The most common presenting symptoms are nonspecific and include weight loss, early satiation, abdominal discomfort, and nausea, many of which were features of this patient's presentation.[12]

In this patient, after a biopsy specimen was obtained with EUS guidance to establish the diagnosis of gastric adenocarcinoma, several staging studies were performed to define the extent of the disease and inform treatment and prognosis. Beyond upper endoscopy and imaging of the chest, abdomen, and pelvis, accurate staging of gastric cancer often requires that diagnostic laparoscopy be performed to examine the peritoneal surfaces and to obtain cytologic washing specimens. Several series have shown that when there is evidence that the tumor has invaded the submucosa or a deeper layer on EUS

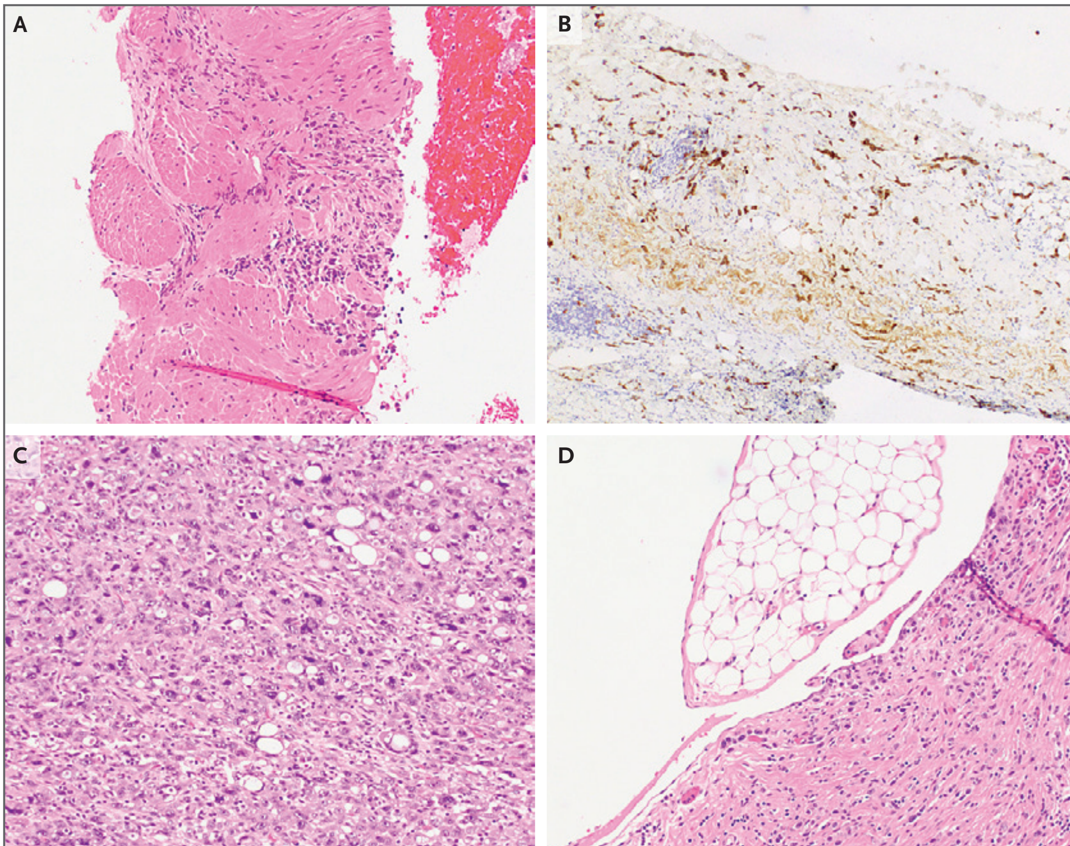

Figure 4. Specimens of the Stomach and Peritoneum.

Hematoxylin and eosin staining of a biopsy specimen from the stomach that was obtained during endoscopic ultrasonography (Panel A) shows fragments of muscularis propria with an associated infiltrative, poorly differentiated adenocarcinoma. Cytokeratin AE1–AE3 and CAM5.2 staining of a biopsy specimen from the peritoneum that was subsequently obtained during laparoscopic evaluation of the abdomen (Panel B) shows infiltrative, poorly cohesive tumor cells in a background of fibroadipose tissue with marked fibrosis and inflammation. Hematoxylin and eosin staining of the gastric-resection specimen (Panels C and D) shows scattered clusters of poorly differentiated adenocarcinoma (Panel C) with extension to the subserosa and serosa (Panel D).

and there is no evidence of metastatic disease on CT, the risk of occult peritoneal disease is 20 to 30%.[13-15] In fact, diffuse gastric cancer may have a tropism for peritoneal spread. To determine the extent of disease in this patient, staging laparoscopy and peritoneal biopsy were performed.

Dr. Judelson: Histologic examination of the biopsy specimen from the peritoneum showed infiltrative, poorly cohesive tumor cells in a background of fibroadipose tissue with marked fibrosis and inflammation. The tumor cells were highlighted on staining for cytokeratin AE1–AE3 and CAM5.2 (Fig. 4B) and for claudin 4. These findings were consistent with metastatic poorly differentiated adenocarcinoma.

Dr. Klempner: Management of advanced gastric cancer involves the balance between maximizing survival and optimizing quality of life. Advanced gastric cancer is associated with a poor prognosis, with the median survival ranging from 11 to 17 months according to recent phase 3 trials.[16,17] Approved therapies can be divided into the following broad categories: cytotoxic agents (chemotherapy), targeted and biologic agents, and immune-checkpoint inhibitors. The selection and combination of these approaches are individualized on the basis of biomarker testing, which is a critical component of managing advanced disease.

Substantial advances in understanding the molecular landscape of gastric cancer have been made, and several molecular classification schemas have been developed.[18-20] Among the recurrent features that are most relevant to the treatment of patients with advanced disease are the overexpression or amplification of the *ERBB2* (*HER2*) gene and corresponding protein, the presence of mismatch-repair defects or microsatellite instability, and the expression of PD-L1. Additional biomarkers of interest include the tumor mutational burden, *NTRK* fusions, claudin 18.2 expression, and *EGFR*, *FGFR2*, and *MET* alterations.[19] This patient's cancer was negative for *HER2* overexpression or amplification, mismatch-repair defects (i.e., he was considered to be mismatch-repair–proficient), and PD-L1 expression on immunohistochemical staining assessed with the use of the combined positive score. The combined positive score is designed to identify patients with gastroesophageal cancers who are most likely to benefit from therapy with an immune-checkpoint inhibitor. A score of 1 or higher is considered to be positive; this patient's score was less than 1.

CYTOTOXIC AGENTS

Platinum-based chemotherapy — which most commonly consists of fluorouracil and oxaliplatin, with possible regimens including FOLFOX (fluorouracil, leucovorin, and oxaliplatin) and CAPOX (capecitabine and oxaliplatin) — is considered to be the standard frontline treatment for advanced gastric cancer. The response rate associated with the use of fluorouracil and oxaliplatin is approximately 45 to 55%.[16,17] Recently, the Food and Drug Administration approved the use of the anti–programmed death 1 antibody nivolumab in combination with chemotherapy as frontline treatment for advanced gastric cancer.[16] However, it is important to note that the magnitude of benefit from the addition of nivolumab is linked to the combined positive score for PD-L1 expression. This patient's cancer was negative for PD-L1 expression, so the benefit from nivolumab would be limited. He was treated with standard FOLFOX chemotherapy.

SURGERY

Although the role for surgical resection in the treatment of advanced gastric cancer is evolving, surgical resection is not currently considered to be a component of standard treatment for patients with stage IV disease. However, given the tropism for peritoneal spread and the symptoms related to the primary tumor, new approaches continue to be explored. In addition, there is some evidence of durable survival after surgery in select patients with stage IV disease.[21]

This patient had refractory symptoms related to the primary tumor, and his case was presented at a multidisciplinary conference for discussion. Multidisciplinary discussion is a central component in the management of many advanced cancers, and shared decision making with patients is important, especially when the approach deviates from standard treatment.[22] For this patient, nausea control and the ability to resume oral food intake were of central importance. After several discussions with the patient, a palliative distal gastrectomy with a retrocolic Billroth II gastrojejunostomy was performed.

Dr. Judelson: Histologic examination of the gastric-resection specimen showed scattered clusters of poorly differentiated adenocarcinoma with extension from the lamina propria to the subserosa and serosa (Fig. 4C and 4D). There was an associated fibrotic response that was consistent with a partial response to chemotherapy. All examined lymph nodes were negative for carcinoma; however, extensive lymphovascular invasion was identified throughout the perigastric adipose tissue. These findings were consistent with a residual poorly differentiated adenocarcinoma, pathologic stage ypT4a N0. This stage indicates a pretreated (y), primary (p) tumor that invades the serosa (T4a) with no positive lymph nodes (N0), according to the American Joint Committee on Cancer staging manual (8th edition).

FOLLOW-UP

Dr. Klempner: After surgery, the patient had a dramatic decrease in nausea and increase in oral intake. He was able to stop receiving enteral nutritional support and to resume physical activities such as wood splitting, martial arts, and regular gym exercise. After completing 3 months of postsurgical FOLFOX chemotherapy, he elected to take a treatment break to optimize his quality of life. At the time of this publication, the patient is not receiving treatment and has no radiographic evidence of disease. He has enjoyed excellent quality of life during the 18 months since the diagnosis of advanced gastric cancer.

FINAL DIAGNOSIS

Linitis plastica (invasive gastric adenocarcinoma).

This case was presented at the Medicine Case Conference.
Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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